

Part 3: Reinforce with Baseline Multi-Trader-Type Crypto Market Analysis

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1 Methodology

1.1 REINFORCE with Baseline

REINFORCE is an on-policy Monte Carlo policy-gradient algorithm. Unlike DQN, which learns a deterministic Q -function off-policy via experience replay, REINFORCE directly parameterises and optimises the stochastic policy $\pi(a | s; \theta)$ through the policy-gradient theorem. The parameter update rule is:

$$\nabla_{\theta} J(\theta) = \mathbb{E} \left[\sum_t \nabla_{\theta} \log \pi(a_t | s_t; \theta) \cdot (G_t - b(s_t)) \right], \quad (1)$$

where $G_t = \sum_{k \geq t} \gamma^{k-t} r_k$ is the discounted return and $b(s_t)$ is a learned baseline — a state-value function $V(s; \phi)$ that reduces gradient variance without introducing bias.

Architecture. Both the policy network and the baseline network are two-layer multilayer perceptrons (MLPs) with ReLU activations and 128 hidden units per layer. Their configurations are summarised in Table 1.

Table 1: Network architectures for the policy and baseline.

Network	Input	Hidden 1	Hidden 2	Output	Activation
Policy $\pi(a s; \theta)$	13-dim state	128, ReLU	128, ReLU	3 logits	Softmax
Baseline $V(s; \phi)$	13-dim state	128, ReLU	128, ReLU	1 scalar	Linear

Training protocol. Each episode spans 168 steps (one trading week). The agent collects a full trajectory under the stochastic policy; at episode end, discounted returns G_t are computed, advantages $A_t = G_t - V(s_t)$ are normalised, and both networks are updated via Adam (lr = 3×10^{-4} , $\gamma = 0.99$, gradient clipping at 1.0). Each trader type is trained for 1,200 episodes on the training set (March 2024 – October 2025).

1.2 Three Trader Types

Three economically distinct agents are constructed by augmenting the base log-return reward

$$r_t = \text{clip} \left(\log \frac{V_t}{V_{t-1}}, -1, 1 \right) \quad (2)$$

with trader-specific incentive terms, as described in Table 2.

1.3 Shapley Value Attribution

Shapley values from cooperative game theory provide the only attribution method satisfying the efficiency, symmetry, dummy, and additivity axioms simultaneously. For each trained policy network, SHAP’s `KernelExplainer` is applied to the buy-action probability $\pi(\text{Buy} | s; \theta)$. `KernelExplainer`

Table 2: Trader types and their reward modifications.

Trader Type		Reward Modification	Economic Rationale
Rational trageur	Arbi-	$r - 0.4 \cdot \sigma_{24} \cdot f_{\text{crypto}}$	Penalises variance of crypto exposure, preferring risk-adjusted returns. Models statistical arbitrageurs that close positions quickly to minimise overnight volatility.
Manipulator (Front-runner)		$r + 0.15 \cdot \bar{V} \cdot \text{sign}(r_{4h}) \cdot d$	Rewards trading in the direction of large-volume moves. Models front-runners who detect institutional order flow via volume spikes and position accordingly. $\bar{V}$: standardised log-volume; $d \in \{+1, 0, -1\}$: buy/hold/sell.
Herd-following Re- tail		$r + 0.25 \cdot \text{sign}(r_{24h}) \cdot d$	Rewards alignment with the 24-hour price trend. Models FOMO-driven retail traders who chase recent momentum regardless of risk.

treats the neural network as a black box, using weighted linear regression over feature coalitions to estimate marginal contributions.

Experimental settings: background set of 30 randomly sampled test states; evaluation over 200 test states with 80 coalition samples per point. Results are reported as mean absolute SHAP values $|\phi_i|$ — the average magnitude of each feature’s contribution to buy-action probability across all test states.

1.4 Market Composition Simulation

To measure systemic effects, a market-composition simulation is run over 500 consecutive test-set hours under seven trader compositions (varying w_{arb} , w_{man} , w_{herd}). At each step:

1. Each agent takes its greedy action given the current market state.
2. An aggregate signal $S_t = w_{\text{arb}} d_{\text{arb}} + w_{\text{man}} d_{\text{man}} + w_{\text{herd}} d_{\text{herd}} \in [-1, +1]$ is computed.
3. Three regime-level metrics are recorded: *crisis frequency*, *price stability* (inverse signal standard deviation), and *market efficiency* ($1 - |\text{lag-1 autocorrelation of } S_t|$).

2 Results

2.1 Backtest Performance

All agents are evaluated on the out-of-sample test period (October 2025 – February 2026), during which BTCB-USD declined 46.2%. Full results are presented in Table 3.

Key finding. The manipulator agent preserved 94% of capital during a 46% market drawdown — the top performer across all methods. Its front-running reward shaped a highly selective buy trigger: the agent enters only when large realised volume and positive four-hour momentum coincide

Table 3: Backtest performance — all agents (test period: Oct 2025 – Feb 2026).

Agent	Method	Return	Sharpe	Max DD	Final \$
Buy & Hold	Passive	−46.21%	−3.201	−49.99%	\$5,379
DQN (Part 2)	Off-policy, det.	−49.44%	−5.301	−53.18%	\$5,056
REINFORCE — Arb.	On-policy, stoch.	−53.07%	−5.424	−56.23%	\$4,693
REINFORCE — Herder	On-policy, stoch.	−17.24%	−1.550	−26.32%	\$8,276
REINFORCE — Manip.	On-policy, stoch.	−6.01%	−3.936	−6.23%	\$9,399

simultaneously, a signal that fires rarely in a bear market. Conversely, the arbitrageur performed worst (−53%): its volatility penalty caused it to hold large positions during brief rallies only to liquidate aggressively as volatility spiked — precisely the wrong sequence in a declining regime. The herder achieved the best Sharpe ratio (−1.55), demonstrating superior risk-efficiency relative to its absolute loss.

REINFORCE vs. DQN. REINFORCE’s stochastic policy proved more adaptive than DQN’s deterministic output. Both the manipulator and herder variants outperformed DQN, which converged to a near-constant hold policy (Q -values: Hold= 0.0043, Buy= 0.0041, Sell= 0.0033). REINFORCE maintains exploration through the policy distribution throughout deployment, whereas DQN’s deterministic arg max collapses to a single action once Q -values converge. Nevertheless, REINFORCE exhibits higher variance: per-episode rewards fluctuate more widely, and the arbitrageur variant underperformed DQN, confirming that *reward design dominates architecture choice*.

2.2 Shapley Value Analysis — Feature Importance by Trader Type

Table 4 reports the top five features for each trader type ranked by mean absolute SHAP value on the buy-action probability.

Table 4: Mean absolute SHAP values for buy-action probability (top 5 features per trader type).

Rank	Arbitrageur		Manipulator		Herder	
	Feature	$ \phi_i $	Feature	$ \phi_i $	Feature	$ \phi_i $
1	rsi_14	0.01595	price_vs_ma50	0.02006	ret_168h	0.01608
2	ret_24h	0.01310	vol_24h	0.01612	price_vs_ma20	0.01188
3	hour_cos	0.01299	price_vs_ma20	0.01437	price_vs_ma50	0.01171
4	ret_168h	0.01243	log_vol	0.01224	rsi_14	0.01122
5	price_vs_ma20	0.01198	hl_range	0.01196	vol_24h	0.01095

Rational Arbitrageur — top features: rsi_14, ret_24h, hour_cos. The arbitrageur’s primary signal is RSI ($|\phi| = 0.01595$), the canonical mean-reversion indicator. At extreme readings (oversold < 30 or overbought > 70) the agent modulates its buying probability, consistent with statistical arbitrage that profits from deviations from equilibrium. The 24-hour return (**ret_24h**) acts as a regime filter: the agent avoids buying into sustained downtrends where mean-reversion has an extended time to materialise. Hour-of-day cosine (**hour_cos**) captures circadian liquidity patterns

— the arbitrageur avoids thin Asian overnight sessions where widened bid–ask spreads amplify the cost imposed by its volatility penalty.

Manipulator — top features: `price_vs_ma50`, `vol_24h`, `price_vs_ma20`. The manipulator’s highest-weight feature is long-term MA deviation (`price_vs_ma50`, $|\phi| = 0.020$). This is economically precise: front-runners use extended price deviations from trend as a signal that a large institutional order is in progress, since large investors typically accumulate or distribute systematically as price recovers toward its trend. The second feature, `vol_24h` ($|\phi| = 0.0161$), acts as a trigger: elevated realised volatility signals active institutional participation and therefore large order flow to precede. `price_vs_ma20` provides shorter-horizon confirmation. Notably, all five top features are market microstructure signals rather than macroeconomic trend indicators, consistent with how real front-runners operate.

Herder — top features: `ret_168h`, `price_vs_ma20`, `price_vs_ma50`. The herder’s dominant feature is the seven-day return (`ret_168h`, $|\phi| = 0.0161$) — the longest-horizon signal in the state space. Retail herders follow trends visible over days or weeks, making the weekly return the clearest signal. The two MA deviation features measure how far price is from trend: herders buy when price sits above its moving average (momentum confirmation) and sell when it falls below. Critically, the herder assigns negligible weight to RSI and volume, indicating that it neither detects overbought/oversold conditions nor interprets institutional order flow. This *absence* of contrarian signals is as informative as what is present.

2.3 Regulatory Implications of Strategic Behaviour

Table 5 summarises the three regime-level metrics under seven trader compositions over the 500-step simulation.

Table 5: Market metrics under different trader compositions (500-step simulation).

Composition	Crisis Frequency	Price Stability	Market Efficiency
All Arbitrageurs	0.362	1.035	0.171
All Manipulators	0.240	1.969	0.338
All Herders	0.018	4.481	0.471
Equal Mix (33/33/34)	0.198	2.798	0.215
Arb-Dominated (60/20/20)	0.362	1.704	0.176
Manip-Dominated (20/60/20)	0.222	2.920	0.307
Herd-Dominated (20/20/60)	0.076	3.937	0.261

Four regulatory insights emerge from these results:

1. **Arbitrageurs reduce efficiency, not crises.** Arbitrageur-dominated markets exhibit the lowest market efficiency (0.171) and the highest crisis frequency (0.362). This counter-intuitive result reflects that arbitrageurs aggressively trade on micro-signals, generating high-frequency signal autocorrelation (reducing efficiency) and actively entering declining regimes (triggering the crisis metric). Regulators should not equate arbitrageur activity with market stability: arbitrageurs stabilise prices locally yet destabilise regimes by amplifying short-term directional signals.

2. **Manipulators are a second-order systemic threat.** Manipulator-dominated markets exhibit moderate crisis frequency (0.240) but notably higher efficiency (0.338) than arbitrageur markets, because manipulators introduce noise that paradoxically breaks autocorrelation. The principal danger from manipulators is wealth extraction — front-running redistributes returns from retail participants to institutions — rather than aggregate systemic instability. This risk is difficult to detect in price-level metrics alone. Regulators must therefore employ trade-level surveillance (clustering of pre-large-order positions) rather than rely exclusively on price monitoring.
3. **Herders create fragility through superficial stability.** Herd-dominated markets appear safest by naive metrics (lowest crisis frequency: 0.018; highest stability: 4.48), yet this apparent calm is misleading. Herders generate predictable, trend-following aggregate signals that are trivially exploitable by manipulators and that can produce sharp reversals when prevailing trends break — the classical bubble-and-bust pattern. Regulators must distinguish between low short-term volatility (a herder characteristic) and genuine market health.
4. **Optimal regulatory composition targets a diversified mixture.** The equal-mix scenario (33/33/34) achieves the most balanced outcome: moderate crisis frequency (0.198), good price stability (2.80), and moderate efficiency (0.215). Regulatory policy should aim to attract arbitrageur participation — through tight spreads, reliable settlement, and the absence of circuit-breaker mechanisms that trap arbitrageurs — while actively surveilling for manipulative behaviour via wash-trading detection and front-running pattern analysis in order books.

3 Conclusions

This study extended the Part 2 BTCB-USD reinforcement learning framework in three directions.

First, REINFORCE with a learned baseline was implemented as a stochastic, on-policy policy-gradient agent. Its distributional policy proved more adaptive than DQN in a bear-market environment: the manipulator-type variant achieved the strongest absolute performance (−6% vs. −46% for buy-and-hold), while DQN converged to a degenerate constant-hold policy.

Second, Shapley-value attribution revealed that the three trader types exploit fundamentally different information channels. Arbitrageurs rely on mean-reversion indicators (`rsi_14`, 24-hour return); manipulators exploit market microstructure signals (long-term MA deviation, realised volatility, volume); and herders weight long-horizon trend signals (7-day return, MA crossovers). These profiles are both economically interpretable and directly actionable for regulators designing automated surveillance systems.

Third, multi-agent market simulation demonstrated that the common assumption that arbitrageurs stabilise markets is incorrect on short-term regime metrics. Manipulator surveillance and herder-behaviour monitoring represent the higher-leverage regulatory interventions for simultaneously preserving market efficiency and price stability.

Implementation: All code available in `rl_crypto_part3.ipynb`. Data: BTCB-USD hourly OHLCV, March 2024 – February 2026. Libraries: PyTorch 2.x, SHAP 0.51, NumPy, Pandas, Matplotlib.